EXPERIMENT:8 POINTERS

Q1.Declare different type of pointers(int,float,char)and initialize them with the addresses of variables.Print the values of both the pointers and the variable they point to.

Algorithm:

Step 1: Start Step 2: Declare variables:

- an integer variable a
- a float variable b
- a char variable c

Step 3: Declare pointers:

- int *p1
- float *p2
- char *p3

Step 4: Initialize pointers with addresses:

- p1 = &a
- p2 = &b
- p3 = &c

Step 5: Print pointer addresses Step 6: Print the values using:

- the variables themselves
- dereferencing the pointers (*p1, *p2, *p3)

Step 7: Stop

code:

```
#include <stdio.h>

int main() {
    int a = 10;
    float b = 20.5;
    char c = 'A';

    int *p1 = &a;
    float *p2 = &b;
    char *p3 = &c;

    printf("----- Pointer Demonstration -----\n");
```

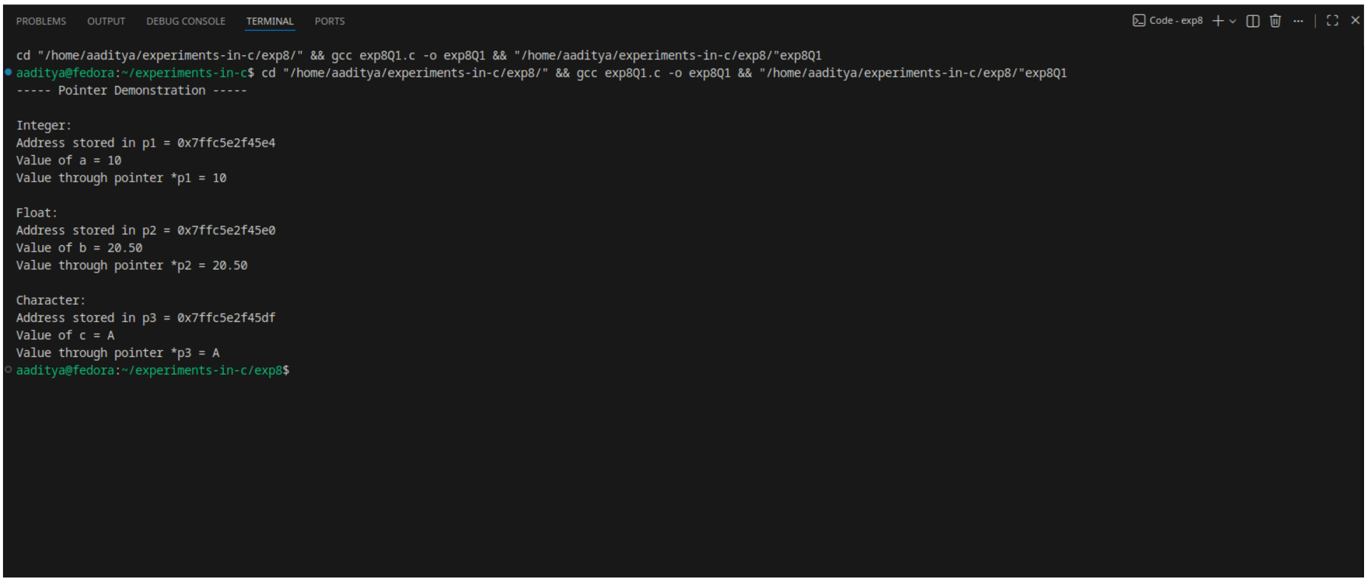
```
    printf("\nInteger:\n");
    printf("Address stored in p1 = %p\n", p1);
    printf("Value of a = %d\n", a);
    printf("Value through pointer *p1 = %d\n", *p1);

    printf("\nFloat:\n");
    printf("Address stored in p2 = %p\n", p2);
    printf("Value of b = %.2f\n", b);
    printf("Value through pointer *p2 = %.2f\n", *p2);

    printf("\nCharacter:\n");
    printf("Address stored in p3 = %p\n", p3);
    printf("Value of c = %c\n", c);
    printf("Value through pointer *p3 = %c\n", *p3);

    return 0;
}
```

Flowchart:

```
              ┌──────────────┐
              │    Start     │
              └──────┬───────┘
                     │
       ┌─────────────▼─────────────┐
       │ Declare a, b, c (variables)│
       └─────────────┬─────────────┘
                     │
       ┌─────────────▼─────────────┐
       │ Declare p1, p2, p3 (pointers)│
       └─────────────┬─────────────┘
                     │
       ┌─────────────▼─────────────┐
       │ p1=&a, p2=&b, p3=&c        │
       └─────────────┬─────────────┘
                     │
       ┌─────────────▼─────────────┐
       │ Print pointer addresses    │
       └─────────────┬─────────────┘
                     │
       ┌─────────────▼─────────────┐
       │ Print values of a, b, c    │
       └─────────────┬─────────────┘
                     │
       ┌─────────────▼─────────────┐
       │ Print *p1, *p2, *p3        │
       └─────────────┬─────────────┘
                     │
              ┌──────▼─────┐
```

```
    |   Stop   |
    └──────────┘
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
cd "/home/aaditya/experiments-in-c/exp8/" && gcc exp8Q1.c -o exp8Q1 && "/home/aaditya/experiments-in-c/exp8/"exp8Q1
aaditya@fedora:~/experiments-in-c$ cd "/home/aaditya/experiments-in-c/exp8/" && gcc exp8Q1.c -o exp8Q1 && "/home/aaditya/experiments-in-c/exp8/"exp8Q1
----- Pointer Demonstration -----

Integer:
Address stored in p1 = 0x7ffc5e2f45e4
Value of a = 10
Value through pointer *p1 = 10

Float:
Address stored in p2 = 0x7ffc5e2f45e0
Value of b = 20.50
Value through pointer *p2 = 20.50

Character:
Address stored in p3 = 0x7ffc5e2f45df
Value of c = A
Value through pointer *p3 = A
aaditya@fedora:~/experiments-in-c/exp8$
```

Q2.Perform pointer arithmetic (increament and decreament)on pointers of different data types.observe how the memory addresses changes and effects on data access.

Algorithm:

Step 1: Start Step 2: Declare variables:

- int a = 10
- float b = 5.5
- char c = 'X'

Step 3: Declare pointers:

- int *p1 = &a
- float *p2 = &b
- char *p3 = &c

Step 4: Print initial pointer values (addresses) Step 5: Perform pointer increment:

- p1++
- p2++
- p3++
- Print new addresses.

Step 6: Perform pointer decrement:

- p1--
- p2--
- p3-- Print new addresses.

Step 7: End Program

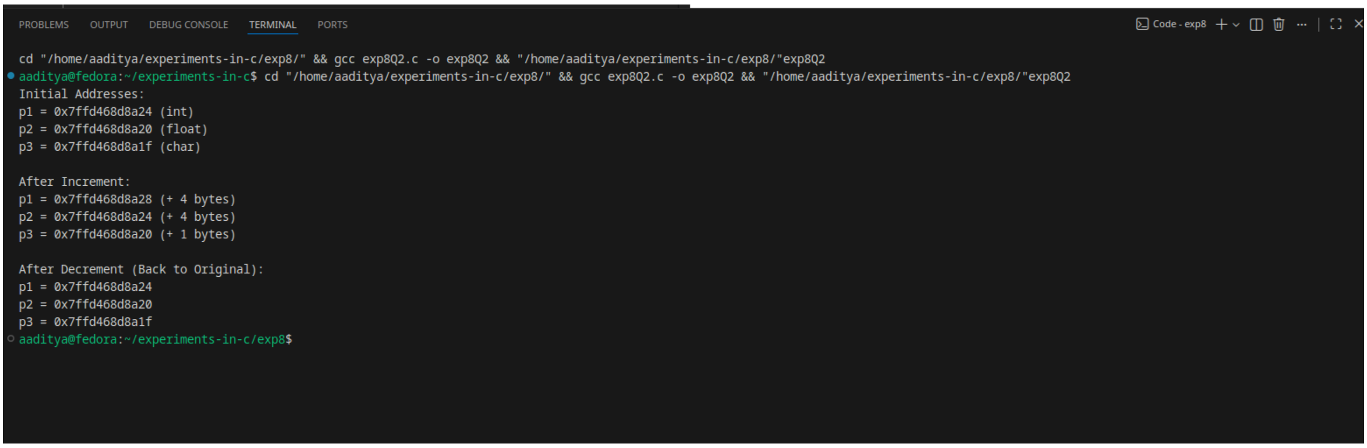
Code:

```
#include <stdio.h>

int main() {
    int a = 10;
    float b = 5.5;
    char c = 'X';

    int *p1 = &a;
    float *p2 = &b;
    char *p3 = &c;

    printf("Initial Addresses:\n");
    printf("p1 = %p (int)\n", p1);
    printf("p2 = %p (float)\n", p2);
    printf("p3 = %p (char)\n", p3);

    // Increment
    p1++;
    p2++;
    p3++;

    printf("\nAfter Increment:\n");
    printf("p1 = %p (+ %lu bytes)\n", p1, sizeof(int));
    printf("p2 = %p (+ %lu bytes)\n", p2, sizeof(float));
    printf("p3 = %p (+ %lu bytes)\n", p3, sizeof(char));

    // Decrement
    p1--;
    p2--;
    p3--;

    printf("\nAfter Decrement (Back to Original):\n");
    printf("p1 = %p\n", p1);
    printf("p2 = %p\n", p2);
    printf("p3 = %p\n", p3);

    return 0;
}
```

Flowchart:

```
┌──────────────┐
│    Start     │
└──────────────┘
       │
       ▼
┌──────────────────────────────┐
│ Declare a, b, c              │
│ Declare p1, p2, p3           │
└──────────────────────────────┘
       │
       ▼
┌──────────────────────────────┐
│ Initialize pointers with     │
│ addresses of a, b, c         │
└──────────────────────────────┘
       │
       ▼
┌───────────────────────────────┐
│ Print initial addresses       │
└───────────────────────────────┘
       │
       ▼
┌───────────────────────────────┐
│ Increment pointers (p++)      │
│ Print new addresses           │
└───────────────────────────────┘
       │
       ▼
┌───────────────────────────────┐
│ Decrement pointers (p--)      │
│ Print addresses               │
└───────────────────────────────┘
       │
       ▼
  ┌─────────────┐
  │    Stop     │
  └─────────────┘
```

Output:



Q3.Write a function that accepts pointers as parameters.pass variables by refrencesusing pointers and modify their values within function.

Algorithm:

Step 1: Start Step 2: Declare two integer variables a and b. Step 3: Read values of a and b from the user. Step 4: Call a function modify(&a, &b) and pass the addresses of variables. Inside modify(x, y):

Add 10 to the value pointed by x

Multiply the value pointed by y by 2

Step 5: Return to main() Step 6: Print modified values of a and b. Step 7: Stop

Code:

```
#include <stdio.h>

// function that modifies values using pointers
void modify(int *x, int *y) {
    *x = *x + 10;   // add 10 to value of x
    *y = *y * 2;    // multiply y by 2
}

int main() {
    int a, b;

    printf("Enter value of a: ");
    scanf("%d", &a);

    printf("Enter value of b: ");
    scanf("%d", &b);

    // calling function with addresses
    modify(&a, &b);

    printf("\nAfter modification:\n");
    printf("a = %d\n", a);
    printf("b = %d\n", b);

    return 0;
}
```

Flowchart:

```
               ┌──────────────┐
               │    Start     │
               └──────┬───────┘
                      │
                      ▼
          ┌─────────────────────────┐
          │ Read a, b from user     │
          └───────────┬─────────────┘
                      │
                      ▼
          ┌─────────────────────────┐
          │ Call modify(&a, &b)     │
```

```
└────────────────────────┘
            │
            │
┌───────────▼────────────┐
│ Print modified a and b  │
└───────────┬────────────┘
            │
            │
      ┌─────▼─────┐
      │   Stop     │
      └───────────┘
```

Output: